Marco Zambon due Fr 20/11/2009

Metodos matematicos da fisica: Homework 5

1 (3 pt)**.** Let $\mathfrak{g}$ be a Lie algebra and $\mathfrak{h}$ a Lie ideal, i.e. a subspace such that $[\mathfrak{h}, \mathfrak{g}] \subset \mathfrak{h}$. Show that the quotient space $\mathfrak{g}/\mathfrak{h}$ admits a unique Lie algebra structure such that the projection map $\pi\colon \mathfrak{g} \to \mathfrak{g}/\mathfrak{h}$ is a Lie algebra morphism.

2 (3 pt)**.** Let $\mathfrak{sl}(2, \mathbb{R})$ be the Lie algebra of all real 2x2-Matrices with trace zero. Show that $\exp : \mathfrak{sl}(2, \mathbb{R}) \to SL(2, \mathbb{R})$ is not surjective.

Hint 1: Relate the (*complex*) eigenvalues of $A \in \mathfrak{sl}(2, \mathbb{R})$ with those of $\exp(A)$.

Hint 2: A different, more computational approach is this. Clearly the following (μ_1, μ_2, μ_3) are a basis of $\mathfrak{sl}(2, \mathbb{R})$:

$$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

One can show that

$$\operatorname{tr}(e^{x_1\mu_1 + x_2\mu_2 + x_3\mu_3}) = 2\cosh\sqrt{|a_1^2 - a_2^2 + a_3^2|},$$

and that there exists $B \in SL(2, \mathbb{R})$ with $\operatorname{tr}(B) < 2$.

3 (3 pt)**.** Let V be a vector space, and for each integer n consider $V^{\otimes n} := V \otimes \cdots \otimes V$. There is a natural action of the symmetric group Σ_n on $V^{\otimes n}$. Define $S^n V$ to be the subspace of $V^{\otimes n}$ on which Σ_n acts by the identity.

a. Show that $S^n V$ is spanned by

$$\{\sum_{\sigma \in \Sigma_n} v_{\sigma(1)} \otimes \cdots \otimes v_{\sigma(n)} : v_1, \ldots, v_n \in V\}.$$

b. Let $A\colon V \to V$ a linear map. Show that the subspace $S^n V$ is invariant under

$$A^{\otimes n}\colon V^{\otimes n} \to V^{\otimes n}, v_1 \otimes \cdots \otimes v_n \mapsto \otimes \cdots \otimes Av_i \otimes \cdots \otimes .$$

c. If $v_i \in V$ is an eigenvector of A with eigenvalue λ_i, $i = 1, \ldots, n$, show that $\sum_{\sigma \in \Sigma_n} v_{\sigma(1)} \otimes \cdots \otimes v_{\sigma(n)}$ is an eigenvector of $A^{\otimes n}$ with eigenvalue $\sum_i \lambda_i$.

d. Let (ρ_3, V_3) denote *the* irreducible complex representation on SU_2 as in class. By ii), there is an induced representation Γ of $SU(2)$ on $S^n V_3$ (obtained restricting the tensor representation on $V^{\otimes n}$). What is the decomposition of Γ into irreducible subrepresentations?

Hint: use iii).